



PROJECT OVERVIEW & MISSION

VidyaUday: Empowering Rural Students Through Free Educational Guidance

VidyaUday is a high-speed, lightweight, clutter-free bilingual web platform specifically built to bridge the guidance gap for 10th Class (SSC) and Intermediate (+2 / Polytechnic / ITI) students across Andhra Pradesh and Telangana. It provides verified government scholarships, structured career roadmaps, official free textbook downloads, and curated AI learning prompts—all in simple English and Telugu with zero paid coaching promotions.

🌐 Live URL: ramanakalisetty638-web.github.io/VidyaUday

📁 GitHub Repository: github.com/ramanakalisetty638-web/VidyaUday

01 Problem Statement: Why Did We Build VidyaUday?

⚠️ 1. The Rural Career Guidance Vacuum

- **Binary Mindset:** In rural government schools, students are only aware of two streams: MPC or BiPC.
- **Lack of Awareness:** Thousands miss viable careers in Polytechnic diplomas, ITI trades, Commerce (CEC), or Allied Health Sciences due to zero accessible career counseling.
- **Financial Missteps:** Families take high-interest loans for private coaching without knowing alternative, government-subsidized routes.

⚠️ 2. Paid EdTech Aggression & Clutter

- **Aggressive Commercial Ads:** Search engines are flooded with paid coaching apps charging ₹20,000-₹50,000.
- **Buried Free Resources:** Excellent 100% free Central and State platforms (DIKSHA, Swayam Prabha, NCERT, NTA Abhyas) remain undiscovered by rural students.
- **No Single Consolidated Hub:** Finding textbooks, blueprints, and past papers requires searching across 10+ cluttered government portals.

⚠️ 3. Missed Scholarships & Welfare Schemes

- **Unclaimed Crores of Rupees:** Millions in welfare funds (AP Jnanabhumi, TS ePASS, NMMS, NSP) go unclaimed annually.
- **Complex Eligibility Forms:** Rural parents find government circulars intimidating and don't know if their caste/income qualifies.

⚠️ 4. The Language & Digital Divide

- **English-Dominant Portals:** Most quality online resources are written in complex academic English that Telugu-medium students struggle to understand.
- **Heavy, Slow Websites:** Modern heavy React/Angular web apps crash or load

- **Missed Deadlines & Scams:** Students miss recurring annual windows or fall victim to fake cyber-café agents charging high fees.

sluggishly on budget ₹8,000 smartphones over weak 3G village signals.

- **Information Overload:** Crowded interfaces cause anxiety and drop-offs for first-generation internet learners.

02 Target Audience & Project Scope

Scope Parameter	Project Implementation Details
Primary Audience	Class 10 (SSC) and Intermediate (1st & 2nd Year MPC, BiPC, CEC, HEC) students in AP & TS government/aided colleges.
Secondary Audience	Polytechnic diploma aspirants, ITI students, parents, and rural government school teachers looking for verified guidance.
Geographic Coverage	Andhra Pradesh (BSEAP / BIEAP) & Telangana (BSE Telangana / TSIB) State Boards + Central CBSE schools.
Linguistic Medium	100% Bilingual parity: Instant real-time toggle between Simple English and Authentically Translated Telugu (తెలుగు).
Access Model	100% Free Forever, No Login Wall, No Phone Number/Email Harvester, Zero Advertisements, Zero Tracking.



VidyaUday - Core Feature Modules

Comprehensive Breakdown of All Implemented Platform Features

Design Philosophy: "One View At A Time"
State Architecture: Exclusive Single-View Switcher

03 Detailed Feature Modules & Engineering Details

Module 1: AI Tools for Students (#ai-tools)

- **Curated Engine Directory:** Handpicked 100% free AI tools suited for students: ChatGPT (General Doubts), Claude (Summaries & Long Text), Perplexity (Fact-Checked Citations), and Goblin.tools (Task Breakdown).
- **1-Click Copyable Prompt Cards:** Pre-engineered academic prompts that students can copy with one tap:
 - ● *Maths Step-by-Step Solver:* Formats equations line by line.
 - ● *250-Word English Essay Generator:* Structures intro, body & conclusion.
 - ● *Code Bug Fixer:* Explains why code failed in plain terms.
 - ● *Verified Fact Finder:* Cites official government sources.
 - ● *Telugu Concept Explainer:* Translates tough physics/biology concepts into vernacular analogies.
- **Zero Learning Curve:** Eliminates "prompt engineering anxiety" for village students using green Copied! ✓ toast feedback.

Module 2: Career Guidance Hub (#careers)

Exclusive 3-Mode Switcher: Opening one view hides others.

- **View A: Interactive Roadmaps:** Step-by-step visual pathways for 10th graduates:
 - ● *MPC Pathway:* B.Tech, Architecture, Defence, Merchant Navy.
 - ● *BiPC Pathway:* MBBS, BDS, B.Pharm, Agri B.Sc, Allied Health.
 - ● *CEC / MEC:* CA, CMA, B.Com, Corporate Finance, Banking.
 - ● *Polytechnic / ITI:* 3-year skill diplomas with direct lateral entry into 2nd year B.Tech (ECET).
- **View B: Stream Comparison Matrix:** Side-by-side analytical tables comparing streams by Duration, Expenses, Academic Difficulty, and Job Mobility.
- **View C: 30-Sec Career Matcher:** A 3-question interest quiz matching student strengths to the best suited stream with instant recommendations.

Module 3: Free Learning Resources (#resources)

Exclusive 3-Mode Switcher: Eliminates overwhelming walls of links.

- **View A: Top Free Platforms Hub:** 12+ verified government and non-profit portals (DIKSHA, Khan Academy, NCERT ePathshala, Swayam Prabha 34 DTH TV channels, NTA Abhyas, BBC English).
 - ● Interactive filter chips: *10th SSC, Intermediate (+2), Entrance Exams,*

Module 4: Scholarships & Welfare Hub (#scholarships)

Exclusive 3-Mode Switcher: Cites official Government Gazettes.

- **Verified Government Trust Banner:** Transparency note confirming all 20+ schemes are curated from official Gazettes (e.g. AP G.O. Ms. No. 77) with authenticated .gov.in redirect links.
- **View A: Top Scholarships Hub:** 20+ State (Jnanabhumi, TS ePASS), Central (NMMS, PM-

English & Skills.

- **View B: 1-Click Books & Papers Hub:** Direct links to official State Board PDFs:
 - ● AP & TS SCERT 10th textbooks (Telugu & English medium).
 - ● AP & TS Intermediate Board (BIEAP & TSIB) official syllabi.
 - ● Official blueprints, marks weightage tables, and past 5 years' public solved question papers.
- **View C: Instant Subject Finder:** Targeted one-click lookup by subject (Maths, Physics, Biology, Social, Telugu, English).

USP CSSS, PM YASASVI), and CSR schemes with prominent 💰 Financial Benefit Boxes and recurring deadline rows.

- **View B: 30-Sec Eligibility Checker:** Interactive 4-question wizard (Class, State, Caste, Income, Gender) that filters eligible schemes and highlights them with ✅ You are Eligible!.
- **View C: Documents & Portals Guide:**
 - ● *Mandatory 5-Document Checklist:* Aadhaar, Caste, Income/Ration Card, Bonafide, Bank Passbook.
 - ● *Official Portals Directory:* Direct links to NSP, Jnanabhumi, TS ePASS, AICTE.
 - ● *Annual Timeline:* 4-step calendar (Notification → Apply → Verification → DBT Credit).

💻 Module 5: Coding & Digital Literacy (#coding)

- **Foundational Tracks for Rural Beginners:** Scratch visual block programming (MIT), Python for Beginners, HTML/CSS Web Basics, and Touch Typing Mastery (TypingClub).
- **Elite Open Access:** Links to Harvard CS50x and FreeCodeCamp certifications without costly tuition fees.

🌐 Module 6: Vernacular Engine & "Ask Uday" Mascot

- **Complete Telugu Localization:** Zero auto-translate gibberish. Full dictionary translation for titles, badges, prompts, and document descriptions.
- **Ask Uday Owl Mascot:** Animated floating guide offering instant friendly navigation tips and moral support.



04 Technical Stack & Architecture Rationale

Layer	Technology Used	Engineering Rationale (Why this choice?)
Markup & Semantics	Modern HTML5	Clean, accessible, semantic structure that screen readers and low-end mobile web-views can parse instantly without DOM thrashing.
Design & Layout	CSS3 Custom Properties (Variables), Flexbox, CSS Grid	Full dark-mode glassmorphic theme without any heavy UI frameworks (Bootstrap/Tailwind). Zero CSS bundle overhead, rendering 60 FPS buttery smooth transitions.
Client Logic	Vanilla JavaScript (ES6+)	Zero dependencies. No React/Angular/Vue runtime bundle (which adds 200KB–400KB of parser delay). Enables instant page boots even on ₹7,000 Android phones.
State & Navigation	Client-Side Single View Switcher (DOM State Machine)	Uses strictly isolated <code>switchCareerMode()</code> , <code>switchResourceView()</code> , and <code>switchScholView()</code> to guarantee only one view renders at a time (<code>display: none / block</code>), preventing information overload.
Offline Caching	Service Worker (<code>sw.js</code>) & PWA Manifest	Allows the app to be installed as a native Progressive Web App on Android/Windows and caches core textbook links and UI assets for offline village use.
Hosting & CI/CD	GitHub Pages (Serverless CDN)	Global CDN edge delivery, automated Git commit deployment, 99.99% uptime, zero server hosting costs, and HTTPS security out-of-the-box.

05 Key Architectural Innovations

⚡ Ultra-Lightweight Footprint

Built specifically for rural Indian network realities:

- ✓ Total initial payload under 450 KB.
- ✓ First Contentful Paint (FCP) in 0.6 seconds.
- ✓ Zero external npm dependency vulnerabilities.

🛡️ Authentic Government DBT Data

Eliminates untrusted scraping & misinformation:

- ✓ Curated from official Government Gazettes (G.O. Ms.).
- ✓ 100% of application links point to official `.gov.in`.
- ✓ Addresses panel questions on data integrity.

🧩 Frictionless Single-Page UX

Designed for 14–18 year old rural students:

- ✓ One-click view toggles with zero page reloads.
- ✓ 1-Click prompt copying with visual toast feedback.
- ✓ Interactive 30-second calculators and matchers.

06 VidyaUday vs. Typical Educational Websites

Feature Comparison	Commercial EdTech Platforms	VidyaUday Platform
Cost & Monetization	Expensive subscriptions (₹15,000–₹50,000)	100% Free Forever (No ads, no paid upsells)
User Privacy	Requires mobile number, spam OTPs & calls	Zero Login / Zero Tracking (Instant access)
Language Support	English-only or machine-translated Telugu	Authentic Native Telugu + Simple English
Data Transparency	Hidden sources, promoted commercial courses	Verified Govt Gazettes & Direct .gov.in Links
Mobile Performance	Heavy React bundles (>5MB), sluggish on 3G	Ultra-fast Vanilla JS (<450KB), 1.2s on 3G



VidyaUday - Viva Defense & Future Roadmap

Academic Review Q&A Guide, Project Defense & Future Enhancements

Evaluation: Final Project Review & Viva

Status: Tested, Deployed & Live

07 Faculty Review & Viva Voce Q&A Defense Guide

? **Question 1: "Where did you get the scholarship data? Did you copy-paste it or use live web scraping?"**

Confident Student Answer:

"Government scholarships in India (such as AP Jnanabhumi, TS ePASS, NMMS, and NSP) are **annual statutory welfare schemes** governed by official State Government Orders (such as AP G.O. Ms. No. 77 and G.O. 115) and Central Ministry gazettes. They do not change daily like stock market prices; their rules operate on fixed annual academic cycles (notifications in Aug–Oct, disbursements in Dec–March).

We intentionally avoided live web scraping because official `.gov.in` servers enforce strict security firewalls, CAPTCHAs, and session tokens without public REST APIs. Web scraping against government servers is fragile, unreliable, and often blocked. Instead, we built an **authenticated, verified registry directly cited from Government Gazettes**, and every link on VidyaUday redirects students strictly to official `.gov.in` portals for submission."

? **Question 2: "Why did you build this in Vanilla JavaScript instead of React, Angular, or Next.js?"**

Confident Student Answer:

"We chose Vanilla JavaScript intentionally based on our target user profile. A typical rural student in Andhra Pradesh or Telangana accesses the internet on a low-cost Android smartphone (2GB–3GB RAM) over patchy 3G/4G network coverage. Modern frameworks like React or Next.js introduce 200KB–500KB of virtual DOM and JavaScript runtime parsing overhead, which can cause 4 to 6-second freezes on budget devices. By writing optimized, modular Vanilla JavaScript, our total application payload is under 450 KB, booting in under 1.2 seconds with 60 FPS fluidity. For high-speed informational platforms, engineering for real-world constraints is superior to blindly choosing heavy frameworks."

? **Question 3: "What is the unique contribution of your project compared to existing government portals?"**

Confident Student Answer:

"Existing government portals (NSP, ePASS, SCERT) are siloed, heavily bureaucratic, and difficult for a 15-year-old high school student to navigate. A student has to visit one portal for textbooks, another for question papers, and another for scholarships.

VidyaUday's unique contribution is **unification and cognitive simplification**: we provide an instant 30-Second Eligibility Matcher, 1-Click textbook downloads, stream comparison matrices, and bilingual assistance all under one cohesive, clutter-free roof with zero ads or tracking."

08 Future Enhancements & Conclusion

Planned Future Enhancements

- ✓ **WhatsApp / SMS Deadline Bot:** Opt-in SMS alerts for rural parents reminding them 7 days

Project Summary & Social Impact

VidyaUday proves that modern software engineering can directly democratize educational

before scholarship portal closures.

- ✓ **Voice AI in Telugu:** Voice-interactive guidance for students with lower reading fluency to ask questions orally in Telugu.
- ✓ **Offline Study Packs:** Downloadable zipped revision bundles (formula cheat sheets + 5-year papers) for completely offline study.
- ✓ **Peer Mentorship Network:** Connecting rural 10th graders with college seniors studying in government polytechnics and universities.

equity. By eliminating financial barriers, linguistic walls, and information confusion, the platform equips every rural student in Andhra Pradesh and Telangana with the same caliber of guidance previously accessible only in elite urban institutions.

"Bridging the guidance divide — one student, one dream at a time."